

Pranshu Gupta

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Professional Experience

Microsoft AI • Senior Software Engineer • Aug 2025 – Present

Build and own the LLM inference layer for Microsoft Copilot, serving consumer and enterprise users globally.

- Reduced production GPU footprint by 1320 H100s by eliminating unnecessary classifier and chain of thought summarization calls from the GPT-5.1 instant inference path, unblocking worldwide model release for consumer copilot.
- Led end-to-end migration of GPT-5 and GPT-5.1 reasoning models to OpenAI's Harmony inference stack, including prompt engineering, constrained sampling, chain of thought handling, safety integration and response quality evals.
- Architected a feature flag system for Harmony API, enabling safe rollouts and A/B experiments for MAI and M365 orgs.
- Unblocked production launch of a new LLM input safety classifier by diagnosing and resolving inference stack integration issues across teams; helped cut the model deployment cycle from >4 months to <1 month.
- Eliminated deployment-time connection errors by implementing session affinity drain and in-memory caching.

Microsoft Azure AI • Senior Software Engineer • Jan 2025 – Aug 2025

Led development and architecture for the data plane APIs (Chat Completions and Responses) serving all LLM inference on Azure.

- Designed a distributed token-based rate limiter using OpenAI's TikToken tokenizer for accurate token counting and response-token quota reconciliation, eliminating a 219% token counting error in the prior estimation-based system.
- Contributed to design and implementation of a prompt-level LLM router that selects the optimal model for each request based on complexity and cost signals, reducing per-inference cost by 55%.

Microsoft Azure Core • Senior Software Engineer • Mar 2024 – Dec 2025

Led the architecture for resource provisioning in Azure Resource Manager, the unified control plane of Azure.

- Designed and shipped execution cost based throttling system for background jobs with multi-bucket enforcement (user, app, subscription, workflow), preventing partition level retry-storm outages and noisy neighbor issues in control plane.
- Patched a critical security vulnerability in Azure control plane; drove adoption across 200+ service teams.

Microsoft Azure Core • Software Engineer II • Jun 2020 – Mar 2024

Designed and developed components for Azure Resource Manager, the unified control plane of Azure.

- Designed and shipped an async callback architecture, reducing polling traffic by up to 94% (about 130 million fewer polling requests/day), while achieving a 12 second reduction in p95 latency for provisioning container instances on Azure.
- Implemented exponential backoff and improved retry patterns for background jobs in Azure Resource Manager, eliminating 700M+ unnecessary executions/day and improving system reliability.
- Owned resource group and bulk deletion APIs end-to-end – implemented dependency resolution, async callbacks, reducing failures by 98% in resource group deletion API, and improving p90 latency by 50% in bulk deletion API.

Microsoft Core Services • Software Engineer • Jun 2017 – Aug 2019

Built a service to enrich customer data from marketing campaigns, enabling better sales opportunities for Microsoft.

- Migrated to serverless architecture, reducing operation costs by 90% and improving performance by 3x.

Technical Skills

- AI and ML : LLM Inference, OpenAI API, Inference Optimization, Safety Orchestration, Response Evaluations.
- Cloud and Infrastructure : Microsoft Azure, .NET Core/Framework, Redis, Azure CLI, Service Bus
- Programming Languages : C#, Python, C++, PowerShell, Kusto QL, HTML/CSS/JS, Markdown.

Education

Georgia Institute of Technology, Atlanta • May 2020 • MSCS (ML specialization)

Indian Institute of Technology, Kanpur • May 2017 • B.Tech. CS

Publications

- Saha, K., Gupta, P. et al. Observer Effect in Social Media Use | Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems - <https://dl.acm.org/doi/10.1145/3613904.3642078>
- El Sherief, M., Saha, K., Gupta, P. et al. Impacts of school shooter drills on the psychological well-being of American K-12 school communities – <https://www.nature.com/articles/s41599-021-00993-6>